SIEVE OF ERATOSTHENES

1	2	3	~~4~~	5	~~6~~	7	~~8~~	~~9~~	~~10~~
11	~~12~~	13	~~14~~	~~15~~	~~16~~	17	~~18~~	19	~~20~~
~~21~~	~~22~~	23	~~24~~	~~25~~	~~26~~	~~27~~	~~28~~	29	~~30~~
31	~~32~~	~~33~~	~~34~~	~~35~~	~~36~~	37	~~38~~	~~39~~	~~40~~
41	~~42~~	43	~~44~~	~~45~~	~~46~~	47	~~48~~	~~49~~	~~50~~
~~51~~	~~52~~	53	~~54~~	~~55~~	~~56~~	~~57~~	~~58~~	59	~~60~~
61	~~62~~	~~63~~	~~64~~	~~65~~	~~66~~	67	~~68~~	~~69~~	~~70~~
71	~~72~~	73	~~74~~	~~75~~	~~76~~	~~77~~	~~78~~	79	~~80~~
~~81~~	~~82~~	83	~~84~~	~~85~~	~~86~~	~~87~~	~~88~~	89	~~90~~
~~91~~	~~92~~	~~93~~	~~94~~	~~95~~	~~96~~	97	~~98~~	~~99~~	~~100~~

From the Sieve of Eratosthenes, we observe that,

- The crossed-out numbers are ***composite*** and the ***coloured numbers*** (encircled) are ***primes.*** The total number of primes upto 100 is 25.
- The only prime number that ends with 5 is 5.

1.2.2 Expressing a Number as the Sum of Prime Numbers

Any number greater than 3 can always be expressed as the sum of two or more prime numbers. Let us illustrate this in the following examples.

Example 1: Express 42 and 100 as the sum of two consecutive primes.

Solution: 42 = 19+23;

100 = 47+53

Example 2: Express 31 and 55 as the sum of any three odd primes.

Solution: 31 = 5+7+19 (find another way, if possible!)

55 = 3 + 23+29



Try these

(i) Express 68 and 128 as the sum of two consecutive primes.

(ii) Express 79 and 104 as the sum of any three primes.

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